

| Question                         |                 |       |  | Marking details                                                                                                                                                                                                                                                                                            | Marks Available |                 |              |       |                      |      |                                  |       |                           |       |   |  |  |   |  |  |
|----------------------------------|-----------------|-------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------|--------------|-------|----------------------|------|----------------------------------|-------|---------------------------|-------|---|--|--|---|--|--|
|                                  |                 |       |  |                                                                                                                                                                                                                                                                                                            | AO1             | AO2             | AO3          | Total | Maths                | Prac |                                  |       |                           |       |   |  |  |   |  |  |
| 5. FT                            | (a)             | (i)   |  | Field B (1)<br>All minerals above <u>threshold</u> value (1)<br>OR comparison to the 2,16, 28                                                                                                                                                                                                              |                 | 2               |              | 2     |                      |      |                                  |       |                           |       |   |  |  |   |  |  |
|                                  |                 | (ii)  |  | Field C (1)<br>Lack of nitrate or potassium (1)                                                                                                                                                                                                                                                            |                 | 2               |              | 2     |                      |      |                                  |       |                           |       |   |  |  |   |  |  |
|                                  |                 | (iii) |  | Add fertiliser                                                                                                                                                                                                                                                                                             | 1               |                 |              | 1     |                      |      |                                  |       |                           |       |   |  |  |   |  |  |
|                                  | (b)             |       |  | <table><tr><th>Argument</th><th>Against (✓) (2)</th></tr><tr><td>cheaper food</td><td></td></tr><tr><td>increased crop yield</td><td></td></tr><tr><td>could cross-pollinate with weeds</td><td>✓ (1)</td></tr><tr><td>long term effects unknown</td><td>✓ (1)</td></tr></table><br>-1 for each extra tick | Argument        | Against (✓) (2) | cheaper food |       | increased crop yield |      | could cross-pollinate with weeds | ✓ (1) | long term effects unknown | ✓ (1) | 2 |  |  | 2 |  |  |
| Argument                         | Against (✓) (2) |       |  |                                                                                                                                                                                                                                                                                                            |                 |                 |              |       |                      |      |                                  |       |                           |       |   |  |  |   |  |  |
| cheaper food                     |                 |       |  |                                                                                                                                                                                                                                                                                                            |                 |                 |              |       |                      |      |                                  |       |                           |       |   |  |  |   |  |  |
| increased crop yield             |                 |       |  |                                                                                                                                                                                                                                                                                                            |                 |                 |              |       |                      |      |                                  |       |                           |       |   |  |  |   |  |  |
| could cross-pollinate with weeds | ✓ (1)           |       |  |                                                                                                                                                                                                                                                                                                            |                 |                 |              |       |                      |      |                                  |       |                           |       |   |  |  |   |  |  |
| long term effects unknown        | ✓ (1)           |       |  |                                                                                                                                                                                                                                                                                                            |                 |                 |              |       |                      |      |                                  |       |                           |       |   |  |  |   |  |  |
|                                  | (c)             |       |  | Any 2 × (1) from:<br>Can cause genetic variation to decrease (1)<br>Some genes would be lost (1)<br>Narrows gene pool (1)<br>Increased risk of genetic disease (1)                                                                                                                                         | 2               |                 |              | 2     |                      |      |                                  |       |                           |       |   |  |  |   |  |  |
|                                  |                 |       |  | Question 5 total                                                                                                                                                                                                                                                                                           | 3               | 4               | 2            | 9     | 2                    |      |                                  |       |                           |       |   |  |  |   |  |  |